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For correspondence:

[aisrael@leumit.co.il](mailto:aisrael@leumit.co.il)

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# Atovaquone/Proguanil Use and Zoster Vaccination Are Associated with Reduced Alzheimer's Disease Risk in Two Cohorts: Implications for a Latent *Toxoplasma gondii* Mechanism

Ariel Israel<sup>1,2</sup>✉, Abraham Weizman<sup>3</sup>, Sarah Israel<sup>4</sup>, Shai Ashkenazi<sup>5,6</sup>, Shlomo Vinker<sup>1</sup>, Eli Magen<sup>7</sup>, Eugene Merzon<sup>1,5</sup>

<sup>1</sup>Leumit Research Institute, Leumit Health Services, Tel Aviv, Israel • <sup>2</sup>Department of Epidemiology and Preventive Medicine, School of Public Health, Faculty of Medical & Health Sciences, Tel Aviv University, Tel Aviv, Israel •

<sup>3</sup>Research Unit, Geha Mental Health Center, Felsenstein Medical Research Center, and Sagol School of Neurosciences, Faculty of Medical & Health Sciences, Tel Aviv University, Tel Aviv, Israel • <sup>4</sup>Department of Clinical Microbiology and Infectious Diseases, Hadassah-Hebrew University Medical Center, Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel • <sup>5</sup>Adelson School of Medicine, Ariel University, Ariel, Israel •

<sup>6</sup>Department of Infectious Diseases, Schneider Children's Medical Center, Petah Tikva, Israel • <sup>7</sup>Medicine A Department, Assuta Ashdod University Medical Center, Ben Gurion University of the Negev, Ashdod, Israel

## eLife Assessment

This **useful** study raises interesting questions but provides **inadequate** evidence of an association between atovaquone-proguanil use (as well as toxoplasmosis seropositivity) and reduced Alzheimer's dementia risk. The findings are intriguing but they are correlative and hypothesis-generating with the strong possibility of residual confounding.

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## Abstract

**Introduction** Identifying modifiable risk factors for Alzheimer's disease (AD) may shed light on novel mechanisms and inform prevention strategies. Increasing evidence suggests that latent pathogens may contribute to AD pathogenesis via chronic neuroinflammation.

**Methods** We conducted a large-scale, dual-cohort study to identify exposures associated with reduced Alzheimer's disease (AD) risk. In a national Israeli cohort (Leumit Health Services; 2004-2024), we analyzed 9,124 AD patients and 18,248 matched controls. We systematically screened medication exposures in the matched cohort for associations with significantly reduced AD risk (OR < 0.5, FDR < 0.05). To account for potential residual confounding, we applied conditional logistic regression models adjusted for age, sex, socioeconomic status, and relevant comorbidities. Findings were independently validated in the U.S.-based TriNetX network, which includes electronic health records from over 120 million patients across 69 healthcare organizations. Propensity score-matched Cox proportional hazards models were used to estimate hazard ratios (HRs) for dementia incidence across stratified age groups.

**Results** Atovaquone/proguanil (Ato/Pro), an antiprotozoal agent active against *Toxoplasma gondii*, was strongly associated with reduced AD risk in both cohorts (LHS: OR 0.36 [95% CI, 0.20-0.61]; TriNetX: HRs 0.34-0.51,  $p = 10^{-17}$  to  $10^{-40}$  across age groups 50-59, 60-69, and 70-79). Both recombinant and live attenuated varicella-zoster virus (VZV) vaccines were also significantly protective (ORs 0.16-0.37), and *T. gondii* seropositivity was associated with a 2.43-fold increased risk of dementia ( $p = 0.0013$ ). Notably, Ato/Pro's protective effect was more pronounced in

individuals without prior VZV vaccination (HR 0.51 [0.43-0.59]) compared to vaccinated individuals (HR 0.71 [0.59-0.85]).

**Discussion** This dual-cohort study - spanning over 120 million patients across two nations - demonstrates strong and reproducible associations linking Ato/Pro use and VZV vaccination to reduced AD risk. The findings support a mechanistic model in which latent *T. gondii* infection, potentially reactivated by herpesvirus co-infection may contribute to AD pathogenesis. Ato/Pro may protect by eliminating or suppressing *T. gondii*, while VZV vaccination may reduce viral triggers of parasite reactivation. These results point to novel preventive strategies and reinforce the infectious hypothesis of Alzheimer's disease.

## Background

Alzheimer's disease (AD) is the most common cause of dementia, characterized by progressive cognitive decline and memory impairment. Globally, over 50 million individuals are affected by dementia, with AD accounting for 60-70% of these cases. This prevalence is projected to triple by 2050, reaching an estimated 152 million cases. (Contador et al., 2024 [↗](#))

The etiology of AD is multifactorial, involving both genetic and environmental factors. Approximately 70% of the risk can be attributed to genetics, notably the presence of the apolipoprotein E ε4 allele. (Silva et al., 2019 [↗](#)) However, modifiable risk factors also play a significant role. Epidemiological studies have identified associations between AD and various factors, including hypertension, lifestyle, and cerebrovascular diseases. Conversely, protective factors such as higher educational attainment, cognitive engagement, and regular physical activity have been linked to a reduced risk of AD. (Qiu et al., 2009 [↗](#)) AD is pathologically characterized by the accumulation of amyloid-beta plaques and neurofibrillary tangles, leading to synaptic dysfunction and neuronal loss. Emerging evidence suggests that pathogen-induced neuroinflammation may exacerbate these pathological features, as infections can trigger chronic inflammatory responses in the brain, potentially accelerating AD progression. (Hensley, 2010 [↗](#))

Recent research has explored the potential protective effects of vaccinations against AD. Immunizations, notably VZV vaccines, have been associated with a decreased risk of developing AD (Pomirchy et al., 2025 [↗](#); Taquet et al., 2024 [↗](#)). The proposed mechanisms include the enhancement of the immune system's ability to combat infections that may contribute to neuroinflammation and subsequent neurodegeneration. (Ukrainitseva et al., 2024 [↗](#)) Vaccination may also modulate immune responses that affect amyloid-beta clearance, reducing AD pathology.

The role of infections in AD pathogenesis and other neurodegenerative processes has garnered increasing attention. Several microbial taxa, including *Porphyromonas gingivalis* (Dominy et al., 2019 [↗](#)), and herpesviruses (Piotrowski et al., 2023 [↗](#); Readhead et al., 2018 [↗](#)), have been implicated in AD pathology through inflammatory and neurotoxic mechanisms. Moreover, antibodies against *Toxoplasma gondii*, a common parasitic infection, have been found at higher rates in individuals with neuropsychiatric disorders such as schizophrenia (Kusbeci et al., 2011 [↗](#); Sutherland et al., 2015 [↗](#); Torrey et al., 2012 [↗](#)), and emerging evidence suggests a possible link to AD (Kusbeci et al., 2011 [↗](#); Wang et al., 2024 [↗](#); Yang et al., 2021 [↗](#)). Chronic infections may contribute to sustained neuroinflammation, leading to neuronal damage, synaptic dysfunction, and cognitive decline. Understanding the interplay between the microbiome, chronic infections, and neurodegeneration is critical for identifying novel preventive and therapeutic strategies for AD. (Lathe et al., 2023 [↗](#))

In this study, we leveraged electronic health records (EHRs) from a national health organization in Israel to systematically investigate whether individuals who developed AD over the past two decades exhibited distinct patterns of previous medication use. Identifying significant reductions in AD occurrence among patients who had received prior antimicrobial treatments may provide insights into the role of targeted microorganisms in the pathogenic processes leading to AD development.

## Methods

### Study design

The first stage of this research was conducted as a retrospective cohort study in Leumit Health Services (LHS), one of the four nationwide health providers in Israel, providing comprehensive healthcare services to approximately 730,000 members. All Israeli citizens are entitled for comprehensive health insurance and receive a standardized package of health services and medications, as defined by the national “Health Basket” committee. LHS operates a centralized EHR system, with over two decades of meticulously maintained information on patient demographics, medical diagnoses, healthcare encounters, laboratory test results, and records of prescribed and purchased medications. Diagnoses are documented during medical encounters by the treating physicians using the International Classification of Diseases (ICD). Diagnoses can be marked as chronic when they pertain to a chronic condition, and these can be updated or closed by the treating physicians during subsequent patient encounters. The reliability of these chronic diagnosis records in our registry has been previously validated, demonstrating high accuracy. (Israel et al., 2021 [↗](#))

Eligibility included all LHS members with active membership between 2003 and 2024. Data were extracted from the LHS central data warehouse, covering diagnoses, laboratory test results, and medication purchases up to December 31, 2024. Structured query language (SQL) and Python scripts automated data retrieval. Socioeconomic status was determined using geocoded residential addresses, classified on a scale of 1 (lowest) to 20 (highest) based on the Points Location Intelligence® database, which strongly correlates with official socioeconomic indicators. Geodemographic classification of the general population, Ultra-Orthodox Jewish, and Arab communities was performed using validated methodologies from prior studies.

### LHS cohort definition

The study included AD patients diagnosed between ages 40 and 89, matched 2:1 with controls who had no dementia diagnosis. To minimize misclassification, individuals with cerebrovascular accident, schizophrenia, Down syndrome, or Parkinson’s disease were excluded. AD cases were identified based on ICD-10 “F00” or ICD-9 “331” codes, with the earliest recorded diagnosis as the index date. Controls were matched to cases by gender, ethnic group (general population, Jewish Ultra-Orthodox, or Arab), socioeconomic status, and year of LHS enrollment. Matching prioritized individuals with the closest birth date to the AD patient, ensuring no duplication.

### TriNetX Validation Cohort

Validation analyses were conducted using TriNetX, a federated research network with EHR data from 69 US healthcare organizations and over 120 million patients. The validation cohorts included individuals aged 65 with no prior dementia diagnosis. Cases were defined as those who had purchased the medication of interest, with controls selected from those who had not. Propensity score matching (PSM) was applied to balance age, ethnicity, race, smoking status, diabetes diagnosis, BMI, and HbA1c levels at medication initiation. Dementia incidence was assessed using Cox proportional hazards models, based on the combined occurrence of ICD-10 codes G30 (Alzheimer’s disease) or F03 (unspecified dementia). Follow-up began 30 days post-index date with no predefined end limit. P-values were calculated using the log-rank test. Using the same methodology, we compared dementia incidence in individuals with past *T. gondii*-positive IgG or IgM serology vs. those with negative serology.

### Data processing

Data from LHS were extracted from EHRs and processed using Python 3.11 scripts and T-SQL queries developed by the Leumit Research Institute. Patient identifiers were removed and replaced with study-specific codes to ensure anonymity. Medication use was assessed based on

purchase records up to 10 years before the index date, classified by Anatomical Therapeutic Chemical (ATC) codes.

## Statistical analyses

Analyses of LHS data were conducted using R version 4.4. Fisher's exact test was used for categorical variables and two-tailed t-tests for continuous variables. Conditional logistic regression models assessed associations between AD and medication use, adjusting for age, gender, smoking status, socioeconomic status, physician visits, pregnancy history, and health worker status.

## Ethics Statement

Ethics approval was granted by the Leumit Health Services Institutional Review Board (LEU-0001-24), with a waiver of informed consent due to retrospective, anonymized data analysis.

## Results

### The study cohort

Table 1 summarizes the demographic and clinical characteristics of 9,124 AD patients and 18,248 controls in the LHS cohort. Due to the matching process, age and gender distributions were very similar (60.7% female, mean age 76.8 ± 8.0 years). AD patients had lower weight (71.5 ± 15.0 vs. 74.3 ± 15.1 kg), height (161 ± 12 vs. 162 ± 10 cm), and BMI. They were more likely to be smokers (OR 1.14 [95% CI, 1.09-1.20]) and had a higher prevalence of diabetes (OR 1.32 [95% CI, 1.25-1.39]), with elevated fasting glucose and hemoglobin A1c levels.

Notable laboratory differences included a higher proportion of AD patients with free T3 levels <3.5 pmol/L (OR 1.42 [95% CI, 1.28-1.58]) or severe vitamin D deficiency of <10 ng/mL (OR 1.32 [95% CI, 1.22-1.44]). Folic acid levels showed a bimodal distribution, with AD patients being more likely to have both deficiency (<2.8 ng/mL, OR 1.48 [95% CI, 1.32-1.65]) and excess (>17 ng/mL, OR 1.30 [95% CI, 1.17-1.43]). Although mean ALT levels were comparable, AD patients had a higher prevalence of very low ALT (<8 U/L, OR 1.83 [95% CI, 1.62-2.07]).

### Medication use associated with the risk of AD occurrence

Medication use over the past decade was systematically assessed for associations with AD risk using Fisher's exact test. Table 2 presents the three medications that met the criteria of OR <0.5 and FDR <5% (Benjamini-Hochberg correction). The two VZV vaccines were significantly associated with lower AD risk: the recombinant VZV vaccine (OR 0.16 [95% CI, 0.05-0.40]) and the live attenuated VZV vaccine (OR 0.37 [95% CI, 0.26-0.52]). Additionally, we observed a significant reduction in AD risk with Atovaquone/Proguanil (Ato/Pro) use, an antiprotozoal used for malaria prophylaxis (OR 0.36 [95% CI, 0.20-0.61]). This combination drug inhibits protozoal mitochondrial function and disrupts folate metabolism.

To further assess the independent effect of each medication on AD risk, we performed an adjusted multivariable regression analysis incorporating these three medications along with potential confounders. The associations remained robust, with adjusted odds ratios and p-values presented in the two rightmost columns of Table 2.

### Validation in the US based trinetX collaborative network

To validate the associations observed in the national cohort, we conducted an independent analysis using TriNetX, a global federated network comprising electronic health records (EHRs) from over 69 U.S. healthcare organizations and more than 120 million patients.

Results are summarized in Table 3, with selected comparisons illustrated as cumulative incidence plots in Figure 1 (subplot letters correspond to the first column of Table 3). The analysis focused on three age strata - 50-59, 60-69, and 70-79 years - each including individuals who had undergone an ambulatory visit during the study period, which served as the index date.

|                           |              | AD<br>N=9,124   | Control<br>N=18,248 | P-<br>value | Odds Ratio       | SMD    |
|---------------------------|--------------|-----------------|---------------------|-------------|------------------|--------|
| Gender                    | Female       | 5,536 (60.7 %)  | 11,072 (60.7 %)     | 1.000       | 1                |        |
|                           | Male         | 3,588 (39.3 %)  | 7,176 (39.3 %)      | 1.000       | 1                |        |
| Age (years)               |              | 76.8 ± 8.0      | 76.7 ± 8.0          | 0.268       |                  | 0.014  |
| Age category              | 40-49        | 93 (1.02 %)     | 184 (1.01 %)        | 0.949       | 1.01 [0.78-1.31] |        |
|                           | 50-59        | 237 (2.60 %)    | 487 (2.67 %)        | 0.749       | 0.97 [0.83-1.14] |        |
|                           | 60-69        | 1,070 (11.73 %) | 2,180 (11.95 %)     | 0.606       | 0.98 [0.90-1.06] |        |
|                           | 70-79        | 3,827 (41.94 %) | 7,729 (42.36 %)     | 0.525       | 0.98 [0.93-1.03] |        |
|                           | 80-89        | 3,897 (42.71 %) | 7,668 (42.02 %)     | 0.048       | 1.02 [1.01-1.15] |        |
| Weight (kg)               |              | 71.5 ± 15.0     | 74.3 ± 15.1         | <0.001      |                  | -0.189 |
|                           | missing      | 73 (2.23 %)     | 2,280 (6.97 %)      |             |                  |        |
| Height (cm)               |              | 161 ± 12        | 162 ± 10            | <0.001      |                  | -0.093 |
|                           | missing      | 122 (3.73 %)    | 3,248 (9.92 %)      |             |                  |        |
| Body Mass Index (kg/m2)   |              | 27.8 ± 5.5      | 28.5 ± 5.2          | <0.001      |                  | -0.131 |
|                           | missing      | 1,775 (19.5 %)  | 2,672 (14.6 %)      |             |                  |        |
| BP systolic (mmHg)        |              | 134 ± 24        | 135 ± 19            | <0.001      |                  | -0.051 |
|                           | missing      | 100 (1.10 %)    | 282 (1.55 %)        |             |                  |        |
| BP diastolic (mmHg)       |              | 75.6 ± 10.1     | 76.0 ± 9.6          | <0.001      |                  | -0.043 |
|                           | missing      | 104 (1.14 %)    | 291 (1.59 %)        |             |                  |        |
| Smoking status            | Non-smoker   | 2,127 (23.6 %)  | 4,564 (25.4 %)      | 0.001       | 0.91 [0.85-0.96] |        |
|                           | Past smoker  | 2,573 (28.5 %)  | 5,396 (30.0 %)      | 0.010       | 0.93 [0.88-0.98] |        |
|                           | Smoker       | 4,320 (47.9 %)  | 7,997 (44.5 %)      | <0.001      | 1.14 [1.09-1.20] |        |
|                           | missing      | 104 (1.1 %)     | 291 (1.6 %)         | 0.003       | 0.71 [0.56-0.89] |        |
| Diabetes Mellitus         |              | 3,335 (36.6 %)  | 5,545 (30.4 %)      | <0.001      | 1.32 [1.25-1.39] |        |
| eGFR MDRD (mL/min/1.73m2) |              | 72.7 ± 26.8     | 72.1 ± 22.8         | 0.080       |                  | 0.023  |
|                           | missing      | 152 (1.67 %)    | 539 (2.95 %)        |             |                  |        |
| Glucose (mg/dL)           |              | 114 ± 42        | 110 ± 31            | <0.001      |                  | 0.122  |
|                           | missing      | 157 (1.72 %)    | 544 (2.98 %)        |             |                  |        |
| Hemoglobin A1c (%)        |              | 6.32 ± 1.32     | 6.19 ± 1.02         | <0.001      |                  | 0.113  |
|                           | missing      | 1,328 (14.6 %)  | 3,382 (18.5 %)      |             |                  |        |
| Free T3 (pmol/L)          |              | 4.03 ± 1.01     | 4.16 ± 0.96         | <0.001      |                  | -0.137 |
|                           | < 3.5 pmol/L | 760 (24.2 %)    | 1,134 (18.4 %)      | <0.001      | 1.42 [1.28-1.58] |        |
|                           | missing      | 6,006 (65.7 %)  | 12,112 (66.2 %)     |             |                  |        |
| Vitamin D (ng/mL)         |              | 21.4 ± 11.0     | 22.4 ± 10.9         | <0.001      |                  | -0.092 |
|                           | < 10 ng/mL   | 1,067 (14.7 %)  | 1,761 (11.5 %)      | <0.001      | 1.32 [1.22-1.44] |        |
|                           | missing      | 1,891 (20.7 %)  | 3,027 (16.6 %)      |             |                  |        |
| Folic Acid (ng/mL)        |              | 8.37 ± 8.37     | 8.28 ± 4.82         | 0.015       |                  | 0.015  |
|                           | < 2.8 ng/mL  | 590 (6.76 %)    | 783 (4.67 %)        | <0.001      | 1.48 [1.32-1.65] |        |
|                           | > 17 ng/mL   | 724 (8.30 %)    | 1,094 (6.53 %)      | <0.001      | 1.30 [1.17-1.43] |        |
|                           | missing      | 418 (4.57 %)    | 1,531 (8.37 %)      |             |                  |        |
| ALT(GPT) (U/L)            |              | 19.2 ± 27.1     | 19.3 ± 17.9         | 0.903       |                  | -0.001 |
|                           | < 8 U/L      | 534 (5.85 %)    | 592 (3.28 %)        | <0.001      | 1.83 [1.62-2.07] |        |
|                           | missing      | 19 (0.21 %)     | 231 (1.26 %)        |             |                  |        |

SMD: Standardized Mean Difference (Cohen's D)

**Table 1.** Demographic and clinical characteristics of the study cohort at index date

| Medication                               | AD<br>N=9,124 | Control<br>N=18,248 | Odds Ratio<br>[95% CI] | adjusted Odds<br>Ratio<br>[95% CI] | adjusted<br>p-value |
|------------------------------------------|---------------|---------------------|------------------------|------------------------------------|---------------------|
| Atovaquone + Proguanil<br>(PO)           | 17 (0.19%)    | 94 (0.51%)          | 0.36 [0.20-0.61]       | 0.41 [0.24-0.69]                   | 0.0008              |
| Zoster, Purified antigen<br>vaccine (IM) | 5 (0.05%)     | 62 (0.34%)          | 0.16 [0.05-0.40]       | 0.20 [0.08-0.50]                   | 0.0006              |
| Zoster, Live attenuated<br>vaccine (SC)  | 41 (0.45%)    | 218 (1.19%)         | 0.37 [0.26-0.52]       | 0.39 [0.28-0.55]                   | <0.0001             |

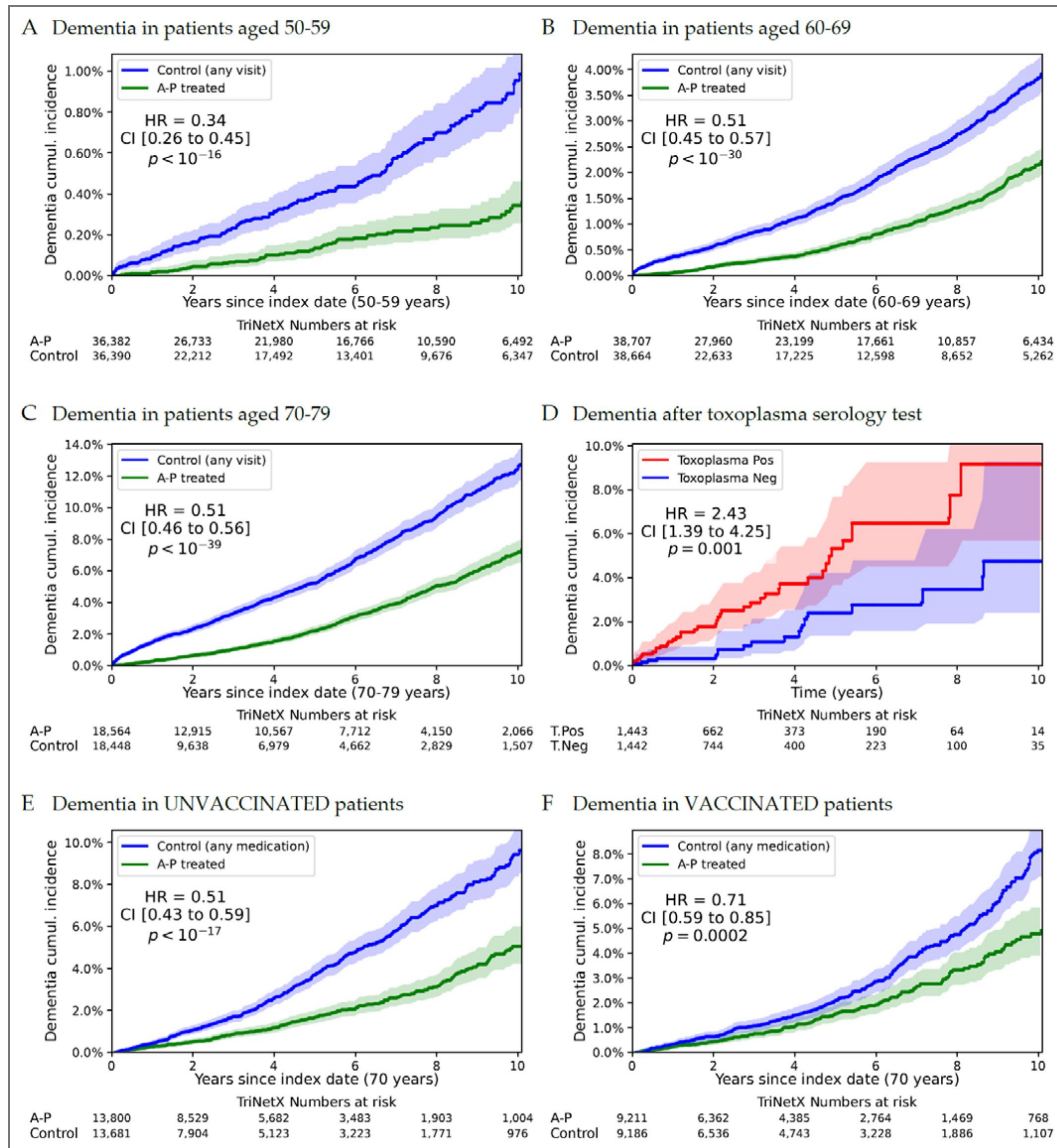
PO: Per Os; SC: Subcutaneous; IM: Intramuscular

Odds ratio were computed according to Fisher's exact tests performed on the matched cohort

Adjusted odds ratio and p-value, were computed according to a conditional logistic regression model which included all the listed medications and is adjusted for age (years), gender, smoking status, SES category, number of past physician visits, previous pregnancy, and health worker status

**Table 2.** Comparison of medication purchases in the 10 years preceding the index date in the AD and control groups of the LHS cohort





**Figure 1. Cumulative incidence of dementia across matched cohorts in TriNetX U.S. data.**

Panels A-C show reduced dementia incidence among individuals previously treated with atovaquone/proguanil (A+P) compared to matched controls across three age groups. Panel D shows increased dementia incidence among *Toxoplasma gondii* seropositive vs. seronegative individuals. Panels E and F stratify the A+P dementia risk by prior varicella-zoster vaccination status. Shaded bands represent 95% confidence intervals. All cohorts were matched 1:1 on age, gender, race/ethnicity, BMI, diabetes, and smoking status. "Control" refers to individuals without A+P exposure (A-C) or with any matched medication use (E, F).

| Sub plot | Group case         | Group control   | Outcome  | Zoster vaccination status | Age   | N matched       | HR   | 95% Confidence Interval | P-value              |
|----------|--------------------|-----------------|----------|---------------------------|-------|-----------------|------|-------------------------|----------------------|
| A        | Atovaquone treated | Any visit       | Dementia |                           | 50-59 | 36,414 / 36,414 | 0.34 | [0.26-0.45]             | $5.9 \cdot 10^{-17}$ |
| B        | Atovaquone treated | Any visit       | Dementia |                           | 60-69 | 38,787 / 38,787 | 0.51 | [0.46-0.57]             | $3.8 \cdot 10^{-31}$ |
| C        | Atovaquone treated | Any visit       | Dementia |                           | 70-79 | 18,676 / 18,676 | 0.51 | [0.46-0.56]             | $1.7 \cdot 10^{-40}$ |
| D        | Toxoplasma pos.    | Toxoplasma neg. | Dementia |                           | 60-75 | 1,456 / 1,456   | 2.43 | [1.39-4.25]             | 0.0013               |
| E        | Atovaquone treated | Any visit       | Dementia | Unvaccinated              | 70    | 13,881 / 13,881 | 0.51 | [0.43-0.59]             | $8.8 \cdot 10^{-18}$ |
| F        | Atovaquone treated | Any visit       | Dementia | Vaccinated                | 70    | 9,258 / 9,258   | 0.71 | [0.59-0.85]             | 0.00022              |

**Table 3. Outcome Comparisons Studies in TriNetX U.S. cohort**

Dementia outcome comparisons in matched cohorts from the TriNetX U.S. network. This table summarizes hazard ratios (HR), confidence intervals, and p-values for dementia incidence across multiple matched comparisons. Subplots A-F correspond to specific analyses displayed in Figure 1. Comparisons include patients exposed to atovaquone/proguanil across three age strata (50-59, 60-69, and 70-79), individuals with positive vs. negative *Toxoplasma gondii* serology (ages 60-75), and stratification by zoster vaccination status at age 70. All cohorts were matched 1:1 by propensity scores on demographics and clinical covariates. Dementia was defined by ICD-10 codes G30 or F03.



The Ato/Pro-exposed groups consisted of individuals with a prior documented purchase of atovaquone and proguanil. Control groups included individuals without such exposure. Propensity score matching was applied to balance the cohorts on age, gender, race/ethnicity, diabetes status, body mass index (BMI), and smoking status. Individuals with a previous diagnosis of Alzheimer's disease or any form of dementia before the index date were excluded from the analysis.

Across all three age strata, Ato/Pro use was consistently associated with a significant reduction in AD risk, closely replicating the findings from our national cohort:

- Ages 50-59: 36,414 matched pairs, HR = 0.34 [95% CI, 0.26-0.45],  $P=5.9 \cdot 10^{-17}$
- ges 60-69: 38,787 matched pairs, HR = 0.51 [95% CI, 0.46-0.57],  $P=3.8 \cdot 10^{-31}$
- Ages 70-79: 18,676 matched pairs, HR = 0.51 [95% CI, 0.46-0.56],  $P=1.7 \cdot 10^{-40}$
- *T. gondii* IgG/IgM positive vs. negative: 1,456 matched pairs, HR = 2.43 [95% CI, 1.39-4.25],  $P=0.0013$

To assess whether *T. gondii*, a parasite eliminated by Ato/Pro, is linked to dementia, we analyzed individuals under 85 years old with a prior *T. gondii* IgG or IgM test. We compared 1,456 individuals with positive serology with a matched group of individuals with negative serology and found a significantly higher incidence of dementia in the *T. gondii* positive group (HR 2.43,  $p=0.001$ ; Fig. 1B [↗](#)). These findings independently confirm the inverse association between Ato/Pro use and AD risk, and support the hypothesis that *T. gondii* may contribute to dementia pathogenesis.

We further examined whether VZV vaccination influenced the protective effect of Ato/Pro by stratifying individuals aged  $\geq 70$  based on their vaccination status. Among unvaccinated individuals (no prior VZV vaccine purchase), Ato/Pro use was associated with a greater and more significant reduction in AD risk (HR 0.51 [95% CI, 0.43-0.59]; Fig. 1C [↗](#)) compared to vaccinated individuals, who had received a VZV vaccine before the index date (HR 0.71 [95% CI, 0.59-0.85]; Fig. 1D [↗](#)). These findings suggest that Ato/Pro's protective effect against AD may be more pronounced in individuals who had not received a VZV vaccine.

## Discussion

This study identifies a novel and robust association between prior atovaquone/proguanil (Ato/Pro) use and a significantly reduced risk of Alzheimer's disease (AD). Conducted within a nationwide health provider over a 20-year period, our large-scale retrospective analysis included 9,124 AD patients and 18,248 matched controls. The inverse association between Ato/Pro use and AD risk was both statistically significant and clinically meaningful, suggesting a previously unrecognized protective factor against AD. Interestingly, our study also identified the protective effect of both the live attenuated and recombinant varicella-zoster virus (VZV) vaccines reported recently by other studies ([Pomirchy et al., 2025 \[↗\]\(#\)](#); [Taquet et al., 2024 \[↗\]\(#\)](#)).

To minimize the possibility of reverse causation, we excluded individuals with any prior diagnosis of dementia or prodromal neurodegenerative conditions at baseline. Additionally, we ensured that all medication exposures occurred at least one month prior to the index date to establish proper temporal sequencing.

Notably, we did not observe a protective association with other medications known to have some anti-*Toxoplasma* activity, such as oral clindamycin. However, this may reflect differences in clinical usage patterns. While Ato/Pro is typically prescribed for extended durations - often spanning several weeks, including pre- and post-travel periods - clindamycin is generally avoided in our health system due to concerns about *Clostridioides difficile* infection risk. When used, it is prescribed almost exclusively to penicillin-allergic patients, typically in short courses and at low doses, which are unlikely to achieve effective anti-*Toxoplasma* activity. This distinction in dosing and treatment context may explain the lack of observed protective effect for clindamycin.

To validate these findings, we independently analyzed data from TriNetX, a U.S.-based global network encompassing electronic health records from over 120 million patients. Using established analytical frameworks and propensity score matching, we observed a consistent and reproducible protective association between Ato/Pro use and reduced dementia risk across three distinct age groups (50-59, 60-69, and 70-79 years). The consistency of this association across age strata, combined with extremely low p-values (ranging from  $10^{-1}$  to  $10^{-}$ ), reinforces the strength and reproducibility of the observed effect.

Importantly, Kaplan-Meier analyses revealed a sustained divergence in cumulative dementia incidence curves extending over more than a decade following Ato/Pro exposure. This long-term protective effect strengthens the case for a causal relationship, with the elimination of a persistent pathogen - most plausibly *Toxoplasma gondii* - representing the most likely underlying mechanism. Consistent with the LHS cohort design, we excluded all individuals with a prior diagnosis of dementia in the TriNetX analyses to minimize confounding by indication and reduce the likelihood of including individuals with early or undiagnosed cognitive decline.

We also found that *Toxoplasma gondii* seropositivity was significantly associated with subsequent dementia development, with most diagnoses occurring many years after the serology was performed. This temporal separation argues strongly against reverse causation - particularly as *T. gondii* serologic testing is not part of routine dementia evaluation - and further supports a potential role for chronic, latent toxoplasmosis in the pathogenesis of Alzheimer's disease.

Interestingly, the protective effect of Ato/Pro was stronger in individuals who had not received a VZV vaccine compared to those who had, suggesting a possible interaction between herpesvirus infection and *T. gondii* reactivation in driving neurodegenerative processes. These observations point to a convergent pathogenic model in which latent *T. gondii* infection, possibly reactivated by herpesvirus superinfection, contributes to chronic neuroinflammation and AD development. In this framework, both Ato/Pro and VZV vaccination act through distinct but complementary mechanisms to reduce AD risk.

Based on our findings, we propose that latent *T. gondii* may contribute to AD pathogenesis. Specifically, we hypothesize that herpesvirus infection may trigger *T. gondii* reactivation, leading to neuroinflammation that drives AD pathology in genetically or immunologically susceptible individuals.

## Toxoplasma gondii

*T. gondii* is a ubiquitous protozoan parasite commonly found in the environment, with humans frequently acquiring asymptomatic, lifelong infections. While generally latent, *T. gondii* has been implicated in a range of neurological and psychiatric disorders (Flegel, 2007 [↗](#); Henriquez et al., 2009 [↗](#)), including schizophrenia (Torrey et al., 2012 [↗](#)), depression (Hlaváčková et al., 2021 [↗](#)), and epilepsy (Ngoungou et al., 2015 [↗](#)). Epidemiological studies have reported associations between *T. gondii* seropositivity and an increased risk of neuropsychiatric conditions, raising the possibility that chronic infection may contribute to neuroinflammation and progressive neuronal damage.

Findings on the association between *T. gondii* and AD have been inconsistent. While some cohorts found no significant link (Wyman et al., 2017 [↗](#)), recent cohort studies reported significant associations between *T. gondii* infection and dementia risk (Wang et al., 2024 [↗](#); Yang et al., 2021 [↗](#)). Additionally, an NHANES analysis found *T. gondii* seropositivity correlated with lower cognitive function scores, further suggesting neurological impacts of chronic infection (Wiener et al., 2020 [↗](#)).

Since *T. gondii* shares mitochondrial and folate metabolism pathways with other protozoa, Atovaquone is effective against it and is used with pyrimethamine as a DHFR inhibitor in toxoplasmosis treatment (KOVACS, 1992 [↗](#)).

Our systematic screening revealed significant AD protection also from both varicella zoster vaccines, likely by preventing herpesvirus infections in neurons, prompting a hypothesis linking herpesvirus reactivation, *T. gondii*, and AD pathology. We propose that in susceptible individuals, herpesvirus infection triggers reactivation of latent *T. gondii*, releasing bradyzoites that could then

spread to adjacent neurons, inducing chronic neuroinflammation contributing to amyloid plaques and neurodegeneration. This is supported by Ato/Pro reducing AD risk much more strongly in patients who had not been vaccinated for VZV (HR=0.51,  $P<0.0001$ ), suggesting both *T. gondii* and herpesvirus (likely VZV) drive pathogenesis. Herpesvirus signatures detected across multiple AD cohorts reinforce this model. (Readhead et al., 2018 [↗](#))

Neuronal cells, being quiescent and metabolically constrained, severely limit *Toxoplasma gondii*'s ability to replicate. Consequently, primary infection is typically contained within the CNS through an immune response that includes microglial activation and cytokine production. This response forces the parasite into its dormant bradyzoite stage, allowing *T. gondii* to evade immune surveillance and persist within host cells. Herpesvirus superinfection of *T. gondii*-harboring cells may disrupt this equilibrium. Viral replication-induced cytolysis can lead to the lysis of infected cells, releasing *T. gondii* into the extracellular space. The resulting neuroinflammatory response—characterized by cytokine release and increased blood-brain barrier (BBB) permeability—may facilitate further invasion of the parasite into adjacent neuronal and glial cells. This cascade of events could amplify neuroinflammation, neuronal damage, and neurodegeneration, ultimately contributing to AD pathology.

## Liver Dysfunction and Immune Dysregulation

A significant proportion of AD patients from the LHS cohort exhibited very low alanine aminotransferase (ALT) levels, suggestive of hepatic dysfunction. The liver plays a key role in immune surveillance and pathogen clearance, filtering pathogens via Kupffer cells, natural killer (NK) cells, and T-cell responses. It also produces complement proteins and cytokines essential for controlling *T. gondii* replication. Interferon-gamma (IFN- $\gamma$ ), in particular, is crucial for host defense, and reduced hepatic cytokine production can impair parasite containment, increasing systemic spread and neuroinvasion.

Based on observed associations in LHS, we hypothesize that liver dysfunction due to frailty, hepatitis, or alcoholism may further compromise these protective mechanisms, enabling *T. gondii* dissemination.

Immune dysregulation in AD patients was also evident in the LHS cohort, with significant reductions in key immune regulators, including vitamin D, T3, and folic acid. The bimodal distribution of folic acid suggests that both deficiency, which can impair immune function, and excessive supplementation, which may promote *T. gondii* replication via its reliance on folic acid metabolism, may influence parasite invasion.

## Conclusion

This large-scale, retrospective study, conducted over 20 years within a national health provider, is the first to identify a strong protective association between Ato/Pro and AD. Validation in the TriNetX U.S. network, encompassing over 120 million patients, reinforces the generalizability of these findings.

The persistence of risk reduction for over a decade post-treatment in both Israeli and American cohorts, alongside increased *T. gondii* seropositivity in AD patients, strongly suggests a link between toxoplasmosis elimination and AD protection. While causality cannot be definitively established, the consistent temporal associations across two large cohorts provide compelling support for the protective effect of Ato/Pro, with *T. gondii* involvement in AD pathogenesis being the most plausible explanation. Moreover, with Ato/Pro's protective effect stronger in VZV-unvaccinated patients, we propose a unifying hypothesis wherein latent *T. gondii* infection, reactivated by herpesvirus infection in the context of immune dysfunction, impaired liver function, and neuroinflammation, ultimately drives AD. Future studies should further explore this mechanism and evaluate targeted interventions. Our findings suggest that controlled *T. gondii* elimination through antiprotazoals, potentially combined with agents that mitigate inflammation and prevent apoptosis in infected cells, such as selective or preferential COX-2 inhibitors, may represent a novel therapeutic avenue. Randomized controlled trials are needed to rigorously test this hypothesis and assess its clinical implications.

## Data Availability

Access to patients data is limited to researchers approved by the Institutional Review Board.

## Additional information

### Funding Sources

This study was funded internally by Leumit Research Institute.

### Consent Statement

Ethics approval was granted by the Leumit Health Services Institutional Review Board (LEU-0001-24), with a waiver of informed consent due to retrospective, anonymized data analysis.

### Author contributions

Conceptualization: AI, EMe, Ema

Methodology: AI, EMe

Investigation: AI, SI, JS, Eme

Writing - original draft: AI

Writing - review & editing: AI, AW, SI, JS, SA, SV, EMa, EMe

### Author ORCID iDs

**Ariel Israel:** <http://orcid.org/0000-0003-4389-8896>

**Abraham Weizman:** <http://orcid.org/0000-0002-9765-8938>

**Shai Ashkenazi:** <http://orcid.org/0000-0001-7244-0679>

**Shlomo Vinker:** <http://orcid.org/0000-0001-9804-7103>

**Eli Magen:** <http://orcid.org/0000-0002-7713-423X>

**Eugene Merzon:** <http://orcid.org/0000-0001-5469-0236>

## References

- Contador I**, Buch-Vicente B, del Ser T, Llamas-Velasco S, Villarejo-Galende A, Benito-León J, Bermejo-Pareja F. (2024) Charting Alzheimer's Disease and Dementia: Epidemiological Insights, Risk Factors and Prevention Pathways. *J Clin Med* **13**:4100 <https://doi.org/10.3390/jcm13144100>
- Dominy SS**, Lynch C, Ermini F, Benedyk M, Marczyk A, Konradi A, Nguyen M, Haditsch U, Raha D, Griffin C, et al. (2019) *Porphyromonas gingivalis* in Alzheimer's disease brains: Evidence for disease causation and treatment with small-molecule inhibitors. *Sci Adv* **5** <https://doi.org/10.1126/sciadv.aau3333>
- Flegr J** (2007) Effects of Toxoplasma on Human Behavior. *Schizophr Bull* **33**:757-760 <https://doi.org/10.1093/schbul/sbl074>
- Henriquez SA**, Brett R, Alexander J, Pratt J, Roberts CW (2009) Neuropsychiatric Disease and *Toxoplasma gondii* Infection. *Neuroimmunomodulation* **16**:122-133 <https://doi.org/10.1159/000180267>
- Hensley K** (2010) Neuroinflammation in Alzheimer's Disease: Mechanisms, Pathologic Consequences, and Potential for Therapeutic Manipulation. *Journal of Alzheimer's Disease* **21**:1-14 <https://doi.org/10.3233/JAD-2010-1414>
- Hlaváčová J**, Flegr J, Fiurašková K, Kaňková Š (2021) Relationship between Latent Toxoplasmosis and Depression in Clients of a Center for Assisted Reproduction. *Pathogens* **10**:1052 <https://doi.org/10.3390/pathogens10081052>
- Israel A**, Merzon E, Schäffer AA, Shenhar Y, Green I, Golan-Cohen A, Ruppín E, Magen E, Vinker S (2021) Elapsed time since BNT162b2 vaccine and risk of SARS-CoV-2 infection: test negative design study. *The BMJ* **375**:e067873 <https://doi.org/10.1136/bmj-2021-067873>

- Kovacs J. (1992) Efficacy of atovaquone in treatment of toxoplasmosis in patients with AIDS. *The Lancet* **340**:637-638 [https://doi.org/10.1016/0140-6736\(92\)92172-C](https://doi.org/10.1016/0140-6736(92)92172-C)
- Kusbeci OY, Miman O, Yaman M, Aktepe OC, Yazar S (2011) Could Toxoplasma gondii Have any Role in Alzheimer Disease?. *Alzheimer Dis Assoc Disord* **25**:1-3 <https://doi.org/10.1097/WAD.0b013e3181f73bc2>
- Lathe R, Schultek NM, Balin BJ, Ehrlich GD, Auber LA, Perry G, Breitschwerdt EB, Corry DB, Doty RL, Rissman RA, et al. (2023) Establishment of a consensus protocol to explore the brain pathobiome in patients with mild cognitive impairment and Alzheimer's disease. *Alzheimer's & Dementia* **19**:5209-5231 <https://doi.org/10.1002/alz.13076>
- Ngoungou EB, Bhalla D, Nzoghe A, Dardé M-L, Preux P-M (2015) Toxoplasmosis and Epilepsy — Systematic Review and Meta Analysis. *PLoS Negl Trop Dis* **9**:e0003525 <https://doi.org/10.1371/journal.pntd.0003525>
- Piotrowski SL, Tucker A, Jacobson S (2023) The elusive role of herpesviruses in Alzheimer's disease: current evidence and future directions. *NeuroImmune Pharmacology and Therapeutics* **2**:253-266 <https://doi.org/10.1515/nipt-2023-0011>
- Pomirchy M, Bommer C, Pradella F, Michalik F, Peters R, Geldsetzer P (2025) Herpes Zoster Vaccination and Dementia Occurrence. *JAMA* <https://doi.org/10.1001/JAMA.2025.5013>
- Qiu C, Kivipelto M, von Strauss E. (2009) Epidemiology of Alzheimer's disease: occurrence, determinants, and strategies toward intervention. *Dialogues Clin Neurosci* **11**:111-128 <https://doi.org/10.31887/DCNS.2009.11.2/cqiu>
- Readhead B, Haure-Mirande J-V, Funk CC, Richards MA, Shannon P, Haroutunian V, Sano M, Liang WS, Beckmann ND, Price ND, et al. (2018) Multiscale Analysis of Independent Alzheimer's Cohorts Finds Disruption of Molecular, Genetic, and Clinical Networks by Human Herpesvirus. *Neuron* **99**:64-82.e7 <https://doi.org/10.1016/j.neuron.2018.05.023>
- Silva MVF, Loures C de MG, Alves LCV, de Souza LC, Borges KBG, Carvalho M das G. (2019) Alzheimer's disease: risk factors and potentially protective measures. *J Biomed Sci* **26**:33 <https://doi.org/10.1186/s12929-019-0524-y>
- Sutherland AL, Fond G, Kuin A, Koeter MWJ, Lutter R, van Gool T, Yolken R, Szoke A, Leboyer M, de Haan L. (2015) Beyond the association. Toxoplasma gondii in schizophrenia, bipolar disorder, and addiction: systematic review and meta analysis. *Acta Psychiatr Scand* **132**:161-179 <https://doi.org/10.1111/acps.12423>
- Taquet M, Dercon Q, Todd JA, Harrison PJ (2024) The recombinant shingles vaccine is associated with lower risk of dementia. *Nat Med* **30**:2777-2781 <https://doi.org/10.1038/s41591-024-03201-5>
- Torrey EF, Bartko JJ, Yolken RH (2012) Toxoplasma gondii and Other Risk Factors for Schizophrenia: An Update. *Schizophr Bull* **38**:642-647 <https://doi.org/10.1093/schbul/sbs043>
- Ukrainitseva S, Yashkin AP, Akushevich I, Arbeev K, Duan H, Gorbunova G, Stallard E, Yashin A (2024) Associations of infections and vaccines with Alzheimer's disease point to a role of compromised immunity rather than specific pathogen in AD. *Exp Gerontol* **190**:112411 <https://doi.org/10.1016/j.exger.2024.112411>
- Wang Jianjun, Lin P, Li D, Yang B, Jiaqi Wang, Feng M, Cheng X (2024) Analysis of the Correlation Between Toxoplasma gondii Seropositivity and Alzheimer's Disease. *Pathogens* **13**:1021 <https://doi.org/10.3390/pathogens13111021>
- Wiener RC, Waters C, Bhandari R (2020) The association of Toxoplasma gondii IgG and cognitive function scores: NHANES 2013-2014. *Parasitol Int* **78**:102123 <https://doi.org/10.1016/j.parint.2020.102123>
- Wyman CP, Gale SD, Hedges-Muncy A, Erickson LD, Wilson E, Hedges DW (2017) Association between Toxoplasma gondii seropositivity and memory function in nondemented older adults. *Neurobiol Aging* **53**:76-82 <https://doi.org/10.1016/j.neurobiolaging.2017.01.018>

Yang H-Y, Chien W-C, Chung C-H, Su R-Y, Lai C-Y, Yang C-C, Tzeng N-S (2021) Risk of dementia in patients with toxoplasmosis: a nationwide, population-based cohort study in Taiwan. *Parasit Vectors* 14:435 <https://doi.org/10.1186/s13071-021-04928-7>

## Peer reviews

### Reviewer #1 (Public review):

#### Summary:

This useful study provides incomplete evidence of an association between atovaquone-proguanil use (as well as toxoplasmosis seropositivity) and reduced Alzheimer's dementia risk. The study reinforces findings that VZ vaccine lowers AD risk and suggests that this vaccine may be an effect modifier of A-P's protective effect. Strengths of the study include two extremely large cohorts, including a massive validation cohort in the US. Statistical analyses are sound, and the effect sizes are significant and meaningful. The CI curves are certainly impressive.

Weaknesses include the inability to control for potentially important confounding variables. In my view, the findings are intriguing but remain correlative / hypothesis generating rather than causative. Significant mechanistic work needs to be done to link interventions which limit the impact of Toxoplasmosis and VZV reactivation on AD.

#### Weaknesses:

#### Major:

(1) Most of the individuals in the study received A-P for malaria prophylaxis as it is not first line for Toxo treatment. Many (probably most) of these individuals were likely to be Toxo negative (~15% seropositive in the US), thereby eliminating a potential benefit of the drug in most people in the cohort. Finally, A-P is not a first line treatment for Toxo because of lower efficacy.

(2) A-P exposure may be a marker of subtle demographic features not captured in the dataset such as wealth allowing for global travel and/or genetic predisposition to AD. This raises my suspicion of correlative rather than casual relationships between A-P exposure and AD reduction. The size of the cohort does not eliminate this issue, but rather narrows confidence intervals around potentially misleading odds ratios which have not been adjusted for the multitude of other variables driving incident AD.

(3) The relationship between herpes virus reactivation and Toxo reactivation seems speculative.

(4) A direct effect on A-P on AD lesions independent on infection is not considered as a hypothesis. Given the limitations above and effects on metabolic pathways, it probably should be. The Toxo hypothesis would be more convincing if the authors could demonstrate an enhanced effect of the drug in Toxo positive individuals without no effect in Toxo negative individuals.

#### Minor:

(5) "Clinically meaningful" should be eliminated from the discussion given that this is correlative evidence.

<https://doi.org/10.7554/eLife.107606.1.sa2>



## Reviewer #2 (Public review):

### Summary:

This manuscript examines the association between atovaquone/proguanil use, zoster vaccination, toxoplasmosis serostatus and Alzheimer's Disease, using 2 databases of claims data. The manuscript is well written and concise. The major concerns about the manuscript center around the indications of atovaquone/proguanil use, which would not typically be active against toxoplasmosis at doses given, and the lack of control for potential confounders in the analysis.

### Strengths:

- (1) Use of 2 databases of claims data.
- (2) Unbiased review of medications associated with AD, which identified zoster vaccination associated with decreased risk of AD, replicating findings from other studies.

### Weaknesses:

- (1) Given that atovaquone/proguanil is likely to be given to a healthy population who is able to travel, concern that there are unmeasured confounders driving the association.
- (2) The dose of atovaquone in atovaquone/proguanil is unlikely to be adequate suppression of toxo (much less for treatment/elimination of toxo), raising questions about the mechanism.
- (3) Unmeasured bias in the small number of people who had toxoplasma serology in the TriNetX cohort.

<https://doi.org/10.7554/eLife.107606.1.sa1>

## Author response:

### Reviewer #1 (Public review):

#### Summary:

*This useful study provides incomplete evidence of an association between atovaquone-proguanil use (as well as toxoplasmosis seropositivity) and reduced Alzheimer's dementia risk. The study reinforces findings that VZ vaccine lowers AD risk and suggests that this vaccine may be an effect modifier of A-P's protective effect. Strengths of the study include two extremely large cohorts, including a massive validation cohort in the US. Statistical analyses are sound, and the effect sizes are significant and meaningful. The CI curves are certainly impressive.*

*Weaknesses include the inability to control for potentially important confounding variables. In my view, the findings are intriguing but remain correlative / hypothesis generating rather than causative. Significant mechanistic work needs to be done to link interventions which limit the impact of Toxoplasmosis and VZV reactivation on AD.*

We thank the reviewer for describing our study as useful and for highlighting several of its strengths, including the very large cohorts, sound statistical analyses, meaningful effect sizes, and the impressive CI curves. We also appreciate the reviewer's recognition that our findings reinforce prior evidence linking VZV vaccination to reduced AD risk.

Regarding the statement that the evidence remains incomplete due to "inability to control for potentially important confounding variables," we refer to our introductory explanation above. As noted there, our analyses meet the accepted criteria for reproducible

epidemiological evidence, and the assumption of uncontrolled confounding is contradicted by rigorous matching and by additional baseline evaluations. We fully agree that mechanistic work is warranted, and our epidemiologic findings strongly motivate such efforts.

We address the reviewer's specific comments in detail below.

*(1) Most of the individuals in the study received A-P for malaria prophylaxis as it is not first line for Toxo treatment. Many (probably most) of these individuals were likely to be Toxo negative (~15% seropositive in the US), thereby eliminating a potential benefit of the drug in most people in the cohort. Finally, A-P is not a first line treatment for Toxo because of lower efficacy.*

We agree that individuals in our cohort received Atovaquone-Proguanil (A-P) for malaria prophylaxis rather than for treatment of toxoplasmosis. However, this does not contradict our interpretation. Because latent CNS colonization by *T. gondii* is not currently considered clinically actionable, asymptomatic carriers are not offered treatment, and therefore would only receive an anti-Toxoplasma regimen unintentionally, through a medication prescribed for another indication such as malaria prophylaxis. Importantly, atovaquone is an established therapy for toxoplasmosis, including CNS disease, with documented efficacy and CNS penetration in current treatment guidelines. It is therefore reasonable to assume that, during the multi-week course typically administered for malaria prophylaxis, A-P would exert significant anti-Toxoplasma activity in individuals with latent CNS infection, potentially reducing or eliminating parasite burden even though the medication was not prescribed for that purpose.

The reviewer notes that only ~15% of individuals in the U.S. are Toxoplasma-seropositive, based on surveys performed primarily in young adults of reproductive age (serologic testing is most commonly obtained in women during prenatal care). However, seropositivity increases cumulatively over the lifespan, and few reliable estimates exist for the age groups in which Alzheimer's disease and dementia occur. Even if we accept the lower estimate of ~15% latent colonization in older adults, this proportion is still smaller than the lifetime cumulative incidence of dementia in the general population.

Therefore, if latent toxoplasmosis contributes causally to dementia risk, and A-P is capable of eliminating latent Toxoplasma in the subset of individuals who harbor it, then a multi-week course of treatment—such as the one routinely taken for malaria prophylaxis—would be expected to produce a substantial reduction in dementia incidence at the population level, of the same order of magnitude reported here. A protective effect concentrated in a minority of exposed individuals is fully compatible with, and can mechanistically explain, the large overall reduction in risk that we observe.

Finally, the reviewer notes that A-P is not a first-line treatment for toxoplasmosis due to assumed lower efficacy. This point does not undermine our results. Even a second-line agent, when administered over several weeks—as is routinely done for malaria prophylaxis—is expected to exert substantial anti-Toxoplasma activity. The long duration of exposure in large populations receiving A-P for travel provides a unique natural experiment that does not exist for other anti-Toxoplasma medications, which, when prescribed for their non-Toxoplasma indications, are not taken more than a few days. Thus, the widespread use of A-P for malaria prophylaxis allows a unique opportunity to evaluate long-term outcomes following inadvertent anti-Toxoplasma treatment.

Moreover, “first line” recommendations in clinical guidelines refer to treatment of acute toxoplasmosis in immunosuppressed individuals, where tachyzoites are actively replicating. These guidelines do not consider efficacy against latent CNS colonization, which is dominated by bradyzoites, a biologically distinct form, in immunocompetent individuals. Therefore, the

guideline hierarchy is not informative regarding which medication is more effective at clearing latent brain infection, the stage we consider most relevant to dementia risk.

*(2) A-P exposure may be a marker of subtle demographic features not captured in the dataset such as wealth allowing for global travel and/or genetic predisposition to AD. This raises my suspicion of correlative rather than casual relationships between A-P exposure and AD reduction. The size of the cohort does not eliminate this issue, but rather narrows confidence intervals around potentially misleading odds ratios which have not been adjusted for the multitude of other variables driving incident AD.*

We agree that prior to matching, A-P exposure may be associated with demographic features such as health or to travel internationally. However, this does not apply after matching. In all age-stratified analyses, exposed and control individuals were rigorously matched on all major risk factors known to influence dementia risk, including age, sex, race/ethnicity, smoking status, hypertension, diabetes, and obesity. Owing to the extremely large pool of individuals in TriNetX (~120M), our matching was performed stringently, producing exposed and unexposed cohorts that are near-identical with respect to the established determinants of dementia risk.

The reviewer correctly identifies that large cohorts alone do not eliminate confounding; however, confounding must still be biologically and epidemiologically plausible. Any hypothetical confounder capable of producing a 50–70% reduction in dementia incidence over a decade would need to: (1) produce a very large protective effect against dementia; (2) be strongly associated with A-P exposure; and (3) remain entirely uncorrelated with age, sex, race/ethnicity, smoking, diabetes, hypertension and obesity, which have been rigorously matched. No such factor has been proposed. The suggestion that an unspecified ‘subtle demographic feature’ could produce effects of this magnitude remains hypothetical, and no such factor has been described in the dementia risk literature.

If a specific evidence-supported confounder is proposed that meets these criteria, we would be pleased to test it empirically in our cohorts. In the absence of such a proposal, the interpretation that the association is merely “correlative rather than causal” remains speculative and does not negate the strength of a replicated, rigorously matched, long-term association across large cohorts in two national health systems.

*(3) The relationship between herpes virus reactivation and Toxo reactivation seems speculative.*

We respectfully disagree with the characterization of the herpesvirus–Toxoplasma interaction as speculative. The mechanism we describe is biologically valid, based on established virology and parasitology literature showing that latent *T. gondii* infection can reactivate from its bradyzoite state under inflammatory or immune-modifying conditions, including viral triggers. A published clinical report has documented CNS co-reactivation of *T. gondii* and a herpesvirus, explicitly noting that HHV-6 reactivation can promote *Toxoplasma* reactivation in neural tissue (Chaupis et al., Int J Infect Dis, 2016).

Moreover, this mechanism is the only currently evidence-supported explanation that simultaneously and parsimoniously accounts for all of the epidemiologic observations in our study:

(1) Substantially higher cumulative incidence of dementia in individuals with positive *Toxoplasma* serology, indicating that latent infection is a risk factor for subsequent cognitive decline;

(2) Strong protective association following A-P exposure, a medication with established activity against *Toxoplasma gondii*, including in the CNS;

(3) Independent protection conferred by VZV vaccination, observed consistently for two vaccines with distinct formulations (one live attenuated, one recombinant protein), whose only shared property is suppression of VZV reactivation;

(4) Greater protective effect of A-P among individuals who were not vaccinated against VZV, consistent with a model in which dementia risk requires both herpesvirus reactivation and persistent latent *Toxoplasma* infection—such that reducing either factor alone (via VZV vaccination or anti-*Toxoplasma* suppression) substantially lowers risk.

Taken together, these observations are difficult to reconcile under any alternative hypothesis.

To date, we are unaware of any other biologically coherent mechanism that can explain all four findings simultaneously. We would welcome any alternative explanation capable of accounting for these converging epidemiologic signals, as such a proposal could meaningfully advance the scientific discussion. In the absence of a competing explanation, the interaction between latent toxoplasmosis and herpesvirus reactivation remains the most parsimonious hypothesis supported by current knowledge.

Finally, while observational studies are inherently limited in their ability to provide causal inference, the mechanism we propose is biologically grounded and experimentally testable. Our results provide a strong rationale for mechanistic studies and clinical trials, and warrant publication precisely because they generate a verifiable hypothesis that can now be evaluated directly.

*(4) A direct effect on A-P on AD lesions independent on infection is not considered as a hypothesis. Given the limitations above and effects on metabolic pathways, it probably should be. The Toxo hypothesis would be more convincing if the authors could demonstrate an enhanced effect of the drug in Toxo positive individuals without no effect in Toxo negative individuals.*

A direct effect of A-P on AD established lesions is indeed possible, and this hypothesis would be of significant therapeutic interest. However, we did not consider it within the scope of our epidemiologic analyses because all cohorts explicitly excluded individuals with existing dementia. Under these conditions, proposing a disease-modifying effect on established Alzheimer's lesions based on our data would itself be speculative. Evaluating such a mechanism would be better answered by mechanistic or interventional studies rather than inference from populations without baseline disease.

We also agree that demonstrating a stronger protective effect among *Toxoplasma*-positive individuals would be informative. Unfortunately, this “natural experiment” cannot be performed using the available data: *Toxoplasma* serology is rarely ordered in older adults, and A-P exposure is itself uncommon, resulting in a cohort overlap far too small to yield valid statistical inference ( $n \approx 25$  in TriNetX).

Thus, while both proposed hypotheses are scientifically attractive and merit further study, neither can be resolved using currently available real-world clinical data. Our findings provide the rationale to investigate both hypotheses experimentally, and we hope our report will motivate such studies.

#### **Reviewer #2 (Public review):**

##### *Summary:*

*This manuscript examines the association between atovaquone/proguanil use, zoster vaccination, toxoplasmosis serostatus and Alzheimer's Disease, using 2 databases of claims data. The manuscript is well written and concise. The major concerns about the manuscript center around the indications of atovaquone/proguanil use, which would not*

*typically be active against toxoplasmosis at doses given, and the lack of control for potential confounders in the analysis.*

*Strengths:*

*(1) Use of 2 databases of claims data.*

*(2) Unbiased review of medications associated with AD, which identified zoster vaccination associated with decreased risk of AD, replicating findings from other studies.*

We thank the reviewer for the thoughtful assessment and for noting key strengths of our work, including (1) the use of two large national databases, and (2) the unbiased discovery approach that replicated the widely reported association between zoster vaccination and reduced Alzheimer's disease (AD) risk. We agree that these features highlight the validity and reproducibility of the analytic framework.

Below we respond to the reviewer's perceived weaknesses.

*Weaknesses:*

*(1) Given that atovaquone/proguanil is likely to be given to a healthy population who is able to travel, concern that there are unmeasured confounders driving the association.*

We agree that, prior to matching, A-P exposure may correlate with demographic or health-related differences (e.g., ability to travel). However, this potential bias was explicitly controlled for in the study design. Across all three age-stratified TriNetX cohorts, exposed and unexposed individuals were rigorously matched on all major established dementia risk factors: age, sex, race/ethnicity, smoking status, obesity, diabetes mellitus, and hypertension. Comparative analyses confirm that these risk factors are equivalently distributed at baseline.

As noted in our response to Reviewer #1, for any hypothetical unmeasured confounder to explain the results, it would need to satisfy three conditions simultaneously:

(1) Be capable of producing a 50–70% reduction in dementia incidence sustained over a decade and across three distinct age strata (ages 50–79);

(2) Be strongly associated with likelihood of receiving A-P;

(3) Remain entirely uncorrelated with age, sex, race/ethnicity, smoking, diabetes, hypertension, or obesity, all of which were rigorously matched and balanced at baseline.

No such factor has been proposed in the literature or by the reviewer. Thus, the concern remains hypothetical and unsupported by any measurable demographic or biological mechanism.

Importantly, empirical evidence contradicts the notion of a “healthy traveler” bias:

Emergency and inpatient encounter rates prior to exposure were comparable between A-P users and controls. Across the three age-stratified cohorts, emergency visits were similar or slightly higher among A-P users (EMER: 19.6% vs 16.4%, 19.9% vs 14.2%, 22.0% vs 14.8%), and inpatient encounters were effectively equivalent (IMP: 14.8% vs 15.2%, 17.7% vs 17.6%, 22.1% vs 22.2%). These patterns directly contradict the suggestion that A-P users were a healthier or less medically burdened population at baseline.

Prevalence of mild cognitive impairment was not lower among A-P users and was, in fact, slightly higher in the oldest cohort. Across the three age groups, baseline diagnoses of mild cognitive impairment (MCI) were comparable or slightly higher among exposed individuals (0.1% vs 0.1%, 0.3% vs 0.2%, 1.1% vs 0.6%). These data contradict the suggestion that A-P users had superior baseline cognition.

The strongest protective association occurred in the youngest stratum (age 50–59; HR 0.34). At this age, when nearly all individuals are sufficiently healthy to travel internationally, A-P uptake is the least likely to confound health status. A frailty-based “healthy traveler” hypothesis would instead predict the opposite pattern, with older adults showing the greatest apparent benefit, since health limitations are more likely to restrict travel in later life. In contrast, the protective association weakens with increasing age, empirically contradicting any explanation based on differential travel capacity.

In conclusion, the empirical evidence directly contradicts the existence of a ‘healthy traveler’ effect.

*(2) The dose of atovaquone in atovaquone/proguanil is unlikely to be adequate suppression of toxo (much less for treatment/elimination of toxo), raising questions about the mechanism.*

A few important points should address the reviewer’s concern:

In our cohorts, A-P was prescribed for malaria prophylaxis, as correctly noted. In this setting, it is taken for the entire duration of travel, plus several days before and after, typically resulting in many weeks of continuous exposure. This creates an unintentional but scientifically valuable natural experiment, in which a CNS-penetrating anti-Toxoplasma agent is administered for long durations.

Atovaquone is an established treatment for CNS toxoplasmosis, has strong CNS penetration, and is included in current clinical guidelines for acute toxoplasmosis in immunocompromised patients, although at higher doses. Because latent, asymptomatic CNS colonization is not treated in clinical practice, there are currently no data establishing the dose required to eliminate bradyzoite-stage Toxoplasma in immunocompetent individuals.

Our observations concern atovaquone–proguanil (A-P), a fixed-dose combination of atovaquone with proguanil, a DHFR inhibitor targeting a key metabolic pathway shared by malaria parasites and *T. gondii*. The combination has well-established synergistic effects in malaria prophylaxis and the same mechanism would be expected to enhance anti-Toxoplasma activity. This fixed-dose regimen has never been formally evaluated for toxoplasmosis treatment at prolonged durations or against latent bradyzoite infection.

Our hypothesis does not require or imply complete eradication of Toxoplasma. A clinically meaningful reduction in latent cyst burden among the subset of colonized individuals may be sufficient to alter long-term disease trajectories. Thus, a population-level decrease in dementia incidence does not require universal clearance of infection, but only partial suppression or reduction of parasite load in susceptible individuals, which is entirely compatible with the known pharmacology and duration of A-P exposure.

*(3) Unmeasured bias in the small number of people who had toxoplasma serology in the TriNetX cohort.*

The relatively small number of older adults with Toxoplasma serology stems from current clinical practice: serologic testing is mostly performed in women during reproductive years due to risks in pregnancy, whereas in older adults a positive result has no clinical consequence and therefore testing is rarely ordered.

Importantly, the seropositive and seronegative groups were drawn from the same underlying population of individuals who underwent serology testing, and the only difference between groups is the test result itself. Because the decision to order a test is made prior to and independent of the result, there is no plausible rationale by which the serology outcome



(positive or negative) would introduce a bias favoring either group beyond the result of the test itself.

Furthermore, the two groups were here also rigorously matched on all major dementia risk factors, including age, sex, race/ethnicity, smoking, diabetes, hypertension, and BMI, and these characteristics are similarly distributed between groups. A small sample size does not imply bias; it simply reduces statistical power. Despite this limitation, the observed association (HR = 2.43,  $p = 0.001$ ) remains strongly significant.

Finally, this result is consistent with multiple published studies reporting higher rates of *Toxoplasma* seropositivity among individuals with Alzheimer's disease, dementia, and even mild cognitive impairment, such that our finding reinforces a broader and independently observed epidemiologic pattern. Importantly, in our cohort the serology testing clearly preceded dementia diagnosis, which supports the plausibility of a causal rather than merely correlative relationship between latent toxoplasmosis and cognitive decline.

To conclude our provisional response, we thank the editor and reviewers for raising points that will be further addressed and expanded upon in the discussion of the forthcoming revision. We welcome transparent scientific dialogue and acknowledge that, as with all observational research, residual confounding cannot be eliminated with absolute certainty. However, we disagree with the overall Assessment and emphasize that our findings—reproduced independently across two national health systems and three age-stratified cohorts, each rigorously matched on all major determinants of dementia risk, meet, and in many respects exceed, current standards for high-quality observational evidence.

Assigning the results to “residual confounding” requires more than speculation: it requires identification of a confounding factor that is (1) anchored in established dementia risk literature, (2) empirically plausible, and (3) quantitatively capable of generating a sustained ~50 percent reduction in dementia incidence over a decade. No such factor has been identified to date. We note that the assertion of “residual confounding” has not been supported by a specific, quantitatively plausible mechanism. A hypothetical bias that is both extremely large in effect and uncorrelated with all major risk factors is not statistically or biologically credible.

The explanation we propose, reduction in dementia risk through elimination of latent *Toxoplasma gondii*, is biologically grounded, directly supported by independent epidemiologic literature, and uniquely capable of accounting for all convergent observations in our data. No alternative hypothesis has been put forward that can plausibly explain these findings.

A revised version of the manuscript will be submitted shortly, incorporating expanded baseline analyses, with the strictest possible exclusion criteria (including congenital, vascular, chromosomal, and neurodegenerative disorders such as Parkinson's disease), and complete tabulated comparisons. These data will further reinforce that the observed protective associations are not attributable to any measurable confounding. We also plan to enhance the discussion in order to address the points raised by the reviewers.

In light of the expanded analyses, any reservations expressed in the initial Assessment can now be re-evaluated on the basis of the empirical evidence. The findings reported in our study meet, and in several respects exceed, current epidemiologic standards for high-quality observational research, clearly warrant publication, and provide a robust scientific foundation for future mechanistic and interventional studies to determine whether elimination of latent toxoplasmosis can prevent or treat dementia.

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